Give the Big O time complexity of the following algorithms

1. Give the time complexity of:

```
None
n = INPUT()
WHILE n > 0:
      OUTPUT(n)
      n = n // 2
ENDWHILE
```

Answer: $O(log\ n)$

2. Give the time complexity of:

```
None
FUNCTION recurse(n):
      IF n <= 1:
            RETURN
      ENDIF
      recurse(n // 2)
      recurse(n // 2)
```

Answer: $O(n)$
Each call does constant work and makes two recursive calls on n/2, so it is linear

3. Give the time complexity of:

```
None
n = INPUT()
FOR i in RANGE(n):
      FOR j in RANGE(i, n):
        OUTPUT(i, j)
      ENDFOR
ENDFOR
```

Answer: $O(n^2)$

4. Give the time complexity of:

```
None
FUNCTION fib(n):
      IF n <= 1:
            return 1
      ENDIF
      fib(n-1) + fib(n-2)
```

Answer: $O(2^n)$